

Check with inverse operations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Check $7,056 \div 24 = 294$. Copy or complete the first step.

2. Check $4,398 + 6,725 = 11,123$.

3. Check $8,402 - 3,765 = 4,637$.

4. Miro says: Miro repeats the same operation and calls it an inverse check. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Check $7,056 \div 24 = 294$. Copy or complete the first step.

Answer $294 \times 24 = 7,056$, so the quotient is correct.

2. Check $4,398 + 6,725 = 11,123$.

Answer $11,123 - 6,725 = 4,398$.

3. Check $8,402 - 3,765 = 4,637$.

Answer $4,637 + 3,765 = 8,402$.

4. Miro says: Miro repeats the same operation and calls it an inverse check. Is this correct? Explain.

Answer Use the opposite operation so the result returns to the starting value.

Check with inverse operations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Check $4,398 + 6,725 = 11,123$.

2. Check $8,402 - 3,765 = 4,637$.

3. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.

4. Identify and correct this misconception: Miro repeats the same operation and calls it an inverse check.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Check $4,398 + 6,725 = 11,123$.
Answer $11,123 - 6,725 = 4,398$.
2. Check $8,402 - 3,765 = 4,637$.
Answer $4,637 + 3,765 = 8,402$.
3. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.
Answer 9,229 because $318 \times 29 + 7 = 9,229$.
4. Identify and correct this misconception: Miro repeats the same operation and calls it an inverse check.
Answer Use the opposite operation so the result returns to the starting value.
5. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Check with inverse operations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.

2. Explain why this reasoning fails: Miro repeats the same operation and calls it an inverse check.

3. Create a new example that tests the same idea as: Check $8,402 - 3,765 = 4,637$.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. A division has quotient 318 remainder 7 and divisor 29. Find the dividend.
Answer 9,229 because $318 \times 29 + 7 = 9,229$.
2. Explain why this reasoning fails: Miro repeats the same operation and calls it an inverse check.
Answer Use the opposite operation so the result returns to the starting value.
3. Create a new example that tests the same idea as: Check $8,402 - 3,765 = 4,637$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.